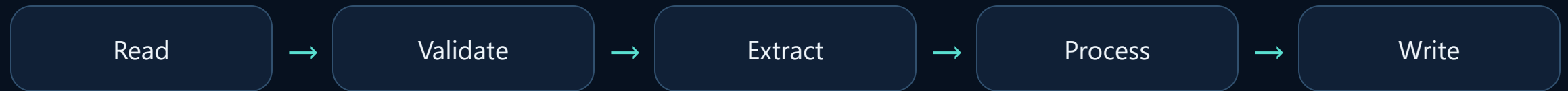


Files and document ingestion.

Bring real documents into a controlled workflow.



**TODAY'S
OUTCOME**

Read, check, extract, and save.

Text

Read UTF-8
safely

PDF

Extract page
text

Failures

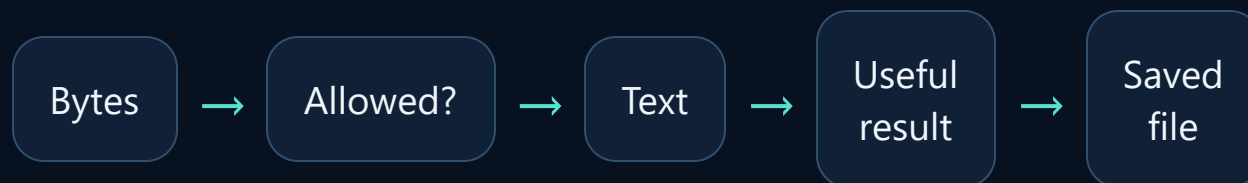
Give clear
actions

Output

Write with
intent

DOCUMENT
PIPELINE

Every stage has a job.



**TRUST BEFORE
TECHNOLOGY**

Can we use this document?

Source

Where did it come from?

Owner

Who controls the data?

Permission

May it leave this machine?

Readable does not mean approved.

PATHS

A path is a file's address.

```
LAB_DIR =  
Path(__file__).resolve().parent.parent  
INPUT_FILE = LAB_DIR / "data" /  
"article.txt"
```

Build from the script location, not the terminal's current folder.

TEXT
FILES

Turn stored bytes into characters.

```
with open(INPUT_FILE, "r",  
          encoding="utf-8") as file:  
    contents = file.read()
```

UTF-8

How bytes
become
characters

with

Closes the file
safely

Reject early with a useful reason.

Exists?

Check the path

Allowed?

Check the
type

Bounded?

Check the size

A PDF is a container.

Text

May be extractable

Images

Need vision or OCR

Layout

May not survive
extraction

SCANNED
DOCUMENTS

Picture of words ≠ stored text.

Text PDF

Select and
extract

Scanned
PDF

OCR may
be
required

OCR output must be
checked for mistakes.

Measure first. Inspect second.

Pages

Did the reader
open them?

Characters

Did text come
back?

Known facts

Is extraction
meaningful?

**MODEL
BOUNDARY**

The document becomes request data.

Privacy

Approved content
only

Context

Text must fit

Instructions

Treat file text as
data

WRITING
OUTPUT

Choose what happens to existing data.

"w"

Create
or
replace

"a"

Add
to
the
end

```
OUTPUT_FILE.parent.mkdir(exist_ok=True)  
OUTPUT_FILE.write_text(result,  
encoding="utf-8")
```

**FAILURE
MAP**

Name the layer that failed.

Input

Missing · type · size

Extract

Encoding · damage · scan

Process

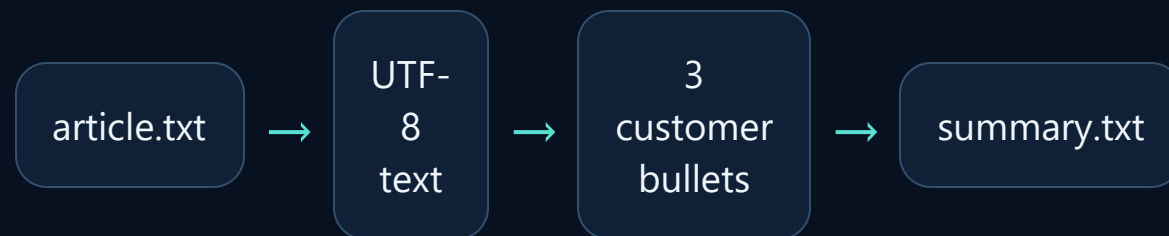
Provider · context

Write

Path · permission · disk

LAB
5.A

Summarize a customer FAQ.



LOCAL
TEXT
PROOF

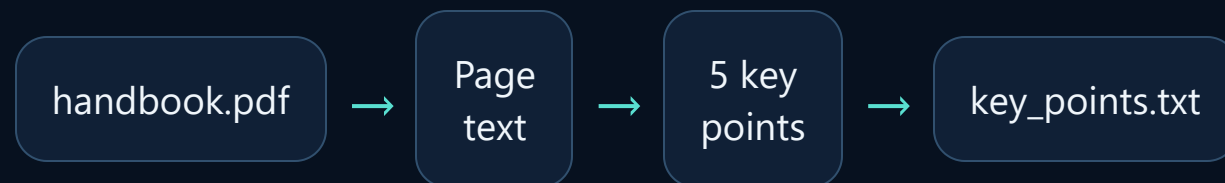
Read the real sample without a provider.

name · bytes ·
characters · safe
first line

Path + decoding +
content presence

LAB
5.B

Extract first-week IT facts.



**LOCAL PDF
PROOF**

Open pages and extract meaningful text.

2 pages

Text found

**Known
facts
visible**

CONTROLLED
PROOF

Fail before the expensive step.

.txt

Accept

.pdf

Accept

.xlsx

Reject
type

missing

Reject path

**PRODUCTION
BOUNDARY**

Untrusted files need strong limits.

Allowlist

Required
formats

Limits

Bytes ·
pages · text

Isolation

Parser and
output

Privacy

Minimal
logs

Day 5 is complete when documents fail clearly.

- ✓ Text and PDF pass local ingestion checks.
- ✓ Unsupported input gets a useful action.
- ✓ Write behavior is understood.
- ✓ Privacy and size limits are recorded.
- ✓ Live-provider status is stated honestly.

Next: Project 1
— Smart
Summarizer.